

nociceptive activation of the trigeminal system in obese mice. Stereotactic radiosurgery (SRS) is used as a noninvasive treatment option for patients with trigeminal neuralgia (TN), but the effect of obesity on pain relief post SRS, if any, is unknown. We hypothesized that obesity would not affect post SRS pain relief.

Materials/Methods: We conducted an institutional review board approved retrospective review of patients treated with single fraction SRS for TN at our institution from 2010-2017. Overweight patients were defined as those with a Body Mass Index (BMI) ≥ 25 . Barrow Neurologic Institute (BNI) Score (1: No pain; 2: Occasional pain; 3: Some pain, adequately controlled with medication; 4: Some pain, not adequately controlled with medication; 5: Severe pain or no relief) was measured pre-and post-SRS to quantify pain level. Improvement in BNI score was defined as the initial (pre-SRS) BNI score subtracted from the final (post-SRS) BNI score. Follow up was defined as time from SRS treatment to last follow up with Neurology or Radiation Oncology, and post-SRS BNI score was taken from this visit.

Results: A total of 34 patients met selection criteria, including 14 (41%) treated for left and 20 (59%) treated for right sided disease; 14 (41%) were men and 20 (59%) women. Five patients had bilateral symptoms. Median prescription dose was 85 Gy (range 75-90), and there was no correlation between overweight status and the SRS dose (Wilcoxon rank sum test $P=0.24$). Median follow up time was 13.3 months (range 3 to 88), and duration of follow up was not associated with overweight status ($p=0.37$). Overweight patients had a lower pre-SRS BNI score (22% vs 0% with BNI score of 0) and had a smaller trigeminal nerve distribution region affected (21% vs 15% affecting one CN V region). Overweight patients were less likely to have constant or lancinating pain (30% vs 47%), and more likely to have multiple sclerosis (9% vs 3%). Overweight patients were less likely to have any improvement in BNI score (63% vs 90% with any reduction in BNI score) and had a smaller improvement (median score improvement of 1 vs 2). Despite poorer response rates, overweight patients were less likely to receive salvage therapy (30% vs 64%). All 3 of the patients who received salvage SRS were overweight. Overweight patients were also more likely to experience any acute or chronic side effect (17% vs 9%). The most common side effect was new facial numbness (13% vs 9%).

Conclusion: While the vast majority (90%) of normal weight TN patients had a reduction in BNI pain score, our study shows that obese patients with TN were less likely to achieve pain relief, more likely to develop side effects, and less likely to be eligible to receive salvage therapies. These results suggest that further investigation is warranted to guide better management for this subgroup of patients.

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The Role of Postoperative Radiation Therapy in the Resectable Desmoid Tumors of the Neck: A Case-Control Study



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Purpose/Objective(s): Desmoid tumors or aggressive fibromatosis are rare soft tissue tumors, which frequently recur after surgery. It is still controversial on the optimal treatment of desmoid tumors because of the lack of large research. Very few studies have included more than 100 patients, especially in one lesion. The purpose of the study was to review a

relative large series of the neck desmoid tumor cases with an emphasis on the value of postoperative radiotherapy.

Materials/Methods: The study group consisted of the neck desmoid tumor patients between 1/2008 and 12/2017, which underwent postoperative radiotherapy at our institution. They were individually matched for a number of factors including age, gender, tumor size and margin status with a cohort of patients who underwent operative without postoperative radiotherapy. The recurrence rates were compared in two groups.

Results: For the study, 34 patients and 34 matched control subjects were identified. The median follow-up time was 38.0 months (range 2-146 months) and 36.5 months (range 5-126 months) in the study groups and the control groups. The two groups did not differ in terms of age ($p = 0.810$), gender ($p = 0.328$), tumor size ($p = 0.803$) and margin status ($p = 0.799$). The interval from surgery to first recurrence in the patients having surgery plus radiotherapy and in those receiving surgery alone was 31.4 ± 25.49 and 13.5 ± 12.98 months, respectively ($p = 0.001$). Patients who received surgery plus radiotherapy had a significantly lower 3-year recurrence rate than those who received surgery alone (25.3% vs. 86.6%, respectively, $P < 0.001$). None of the patients treated with radiotherapy developed serious complications.

Conclusion: According to our results, postoperative radiotherapy is especially effective and necessary in the resectable desmoid tumors of the neck. Postoperative radiotherapy appears to be significantly associated to local control and the complications are acceptable.

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Long-Term Tumor Control with Radiation Therapy for Symptomatic Spine Hemangioma



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Purpose/Objective(s): Data are scarce on long-term results of radiotherapy for symptomatic hemangioma of the spine. The purpose of our study is to report imaging and clinical evidence of tumor control with long-term follow-up after radiotherapy for visible tumor that was not well suited to surgical or interventional radiology treatment.

Materials/Methods: Our study population is all 10 patients treated at our institution between 1968 and 2008 for symptomatic hemangioma of the spine. All patients in our series had substantial pain from visible hemangioma at the time of radiotherapy for which surgical resection or interventional radiology procedures were likely to result in high morbidity. Tumor location in the spine included lumbar in 40%, thoracic in 50%, and cervical in 10% of patients. The mean dose delivered was 47 Gy. Clinical follow-up was defined as no symptoms of recurrent tumor after RT. Imaging follow-up was defined as no increase in tumor size after RT.

Results: The mean imaging follow-up was 8.1 years and the mean clinical follow-up was 21.2 years. The crude calculation rate of local tumor control was 90% (9/10). The only patient with recurrence demonstrated tumor progression 30 years after RT. The actuarial rate of tumor control was 100% at 5, 10, 20, and 25 years. There were no grade 3+ treatment toxicities, no evidence of malignant transformation, and no evidence of second tumors in the treatment area.

Conclusion: RT for symptomatic hemangioma of the spine provides long-term tumor control without complications. Radiation therapy should be considered in patients for whom surgical resection would result in unacceptable morbidity. Our preferred dose program is 45 Gy at 1.8 Gy per treatment.

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